



Drawn from Fuhua Secondary's **2024 (3NA)** and **2025 (3E)** End-of-Year papers. This is the **construction grammar** — the habits that repeat regardless of which topic is being tested. Learn the grammar and the paper stops being a surprise.

1 Structural signature

ELEMENT	FUHUA'S HABIT
Opener	A 2–4 mark algebra warm-up. Quadratic inequality (2025 Q1a), $4x^2 < x$ (2024 Q3), principal value (2024 Q1a). Deliberately low-stakes.
Discriminant	Appears twice , in two disguises: once as a <i>proof</i> ("show ... for all real k "), once as a <i>range</i> or <i>count of intersections</i> . 2025 Q1b + Q2; 2024 Q6 + Q8.
Spine	One 10–12 mark coordinate-geometry question, headed " Solutions by accurate drawing will not be accepted ".
Closer	The trigonometric graph — amplitude, period, sketch. Nearly identical two years running.
Ladder	1–2 mark openers → 4–6 mark middle → one 10–12 mark spine.

2 The five recurring construction devices

1 A prose-reasoning mark in every paper (~2 marks). Not computation — explanation.

- 2025 Q5(i): "*David claims ... Explain why his claim is wrong.*"
- 2024 Q2(ii): "*Explain why $y = -7$ is the only real root.*"
- 2024 Q15(b): "*Explain why $y = 1$ is not a tangent.*"

2 A "for all values of" proof. They want the discriminant shown *positive* or *negative*, with justification — not merely non-zero. 2025 Q1(b).

3 A context-ceiling answer. "*the number of complete hours*" (2025 Q4d). The decimal is not the answer; the next whole unit is.

4 Deliberately ugly exact values. $B(28/9, 4/3)$ in 2025 Q9(iii). The method is standard; the arithmetic is the filter.

5 Answer-form constraints, always stated. "*in the form $a + b\sqrt{3}$ where a and b are integers*" / "*rational numbers*" / "*correct to 2 decimal places*" / "*in radians as a multiple of π* " / "*exact form*". A correct number in the wrong form loses the A mark.

3 Topic frame (Express)

Quadratics & discriminant · **surds** (always *both* rationalising *and* a surd equation) · **polynomials** (remainder / factor → solve cubic) · **partial fractions** (*always* with an irreducible quadratic factor) · **indices & logarithms** (including change of base) · **exponential growth model** · **binomial theorem** · **coordinate geometry** · **circles** · **linear law** · **trigonometric ratios** with negative / complementary angles · **trigonometric graphs**.